



Mark Scheme

Mock set 3

Pearson Edexcel GCE

In Mathematics (8MA0_21) Statistics

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General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

EDEXCEL GCE MATHEMATICS

General Instructions for Marking

1. The total number of marks for the paper is 30.
2. The Edexcel Mathematics mark schemes use the following types of marks:
 - **M** marks: method marks are awarded for 'knowing a method and attempting to apply it', unless otherwise indicated.
 - **A** marks: Accuracy marks can only be awarded if the relevant method (M) marks have been earned.
 - **B** marks are unconditional accuracy marks (independent of M marks)
 - Marks should not be subdivided.

3. Abbreviations

These are some of the traditional marking abbreviations that will appear in the mark schemes.

- bod – benefit of doubt
 - ft – follow through
 - the symbol $\checkmark$ will be used for correct ft
 - cao – correct answer only
 - cso - correct solution only. There must be no errors in this part of the question to obtain this mark
 - isw – ignore subsequent working
 - awrt – answers which round to
 - SC: special case
 - oe – or equivalent (and appropriate)
 - dep – dependent
 - indep – independent
 - dp decimal places
 - sf significant figures
 - * The answer is printed on the paper
 - $\square$ The second mark is dependent on gaining the first mark
4. For misreading which does not alter the character of a question or materially simplify it, deduct two from any A or B marks gained, in that part of the question affected.
 5. Where a candidate has made multiple responses and indicates which response they wish to submit, examiners should mark this response.
If there are several attempts at a question which have not been crossed out, examiners should mark the final answer which is the answer that is the most complete.
 6. Ignore wrong working or incorrect statements following a correct answer.
 7. Mark schemes will firstly show the solution judged to be the most common response expected from candidates. Where appropriate, alternative answers are provided in the notes. If examiners are not sure if an answer is acceptable, they will check the mark scheme to see if an alternative answer is given for the method used.

Question	Scheme		Marks	AOs
1(a)	$X \sim B(10, 0.8)$		B1	3.3
			(1)	
(b)(i) (ii)	$P(X = 8) = 0.30198\dots$ awrt 0.302		B1	1.1b
	$P(X \geq 7) = 1 - P(X \leq 6) = 1 - 0.12087$		M1	3.4
	$= 0.87913\dots$ awrt 0.879		A1	1.1b
			(3)	
(c)	The binomial distribution requires the events to be independent, however, if whales are seen in an area on one day, there may be a higher likelihood that they will be there on the next day o.e.		B1	3.5b
			(1)	
(Total 5 marks)				
Notes:				
(a)	B1	For sight of the correct model $B(10, 0.8)$		
(b)(i) (ii)	B1	awrt 0.302		
	M1	Sight of correct process to find $P(X \geq 7)$, can be implied by statement or correct numbers, allow $1 - 0.121$ or better.		
	A1	awrt 0.879		
(c)	B1	Allow statements that imply that the event of seeing the whales on consecutive days is not independent, or that the probability may not be constant.		

Question	Scheme		Marks	AOs
2(a)	$H_0 : p = 0.4 \quad H_1 : p \neq 0.4$		B1	2.5
	$[X \sim] B(60, 0.4)$		M1	3.3
	$P(X \geq 31) = 1 - P(X \leq 30) (= 1 - 0.9555\dots)$ $= 0.0444803\dots$ awrt 0.0445		A1	3.4
	[“0.0445” > 0.025, therefore not significant, so...] Do not reject H_0 . Not enough evidence to suggest that training with the new psychologist has <u>changed</u> the <u>proportion</u> of her <u>free throws</u> from which she <u>scores</u> .		A1	2.2b
			(4)	
(b)	p -value is $2 \times \text{“0.0444803”} = 0.08896\dots$ awrt 0.089		B1ft	1.1b
			(1)	
(Total 5 marks)				
Notes:				
(a)	B1	For correctly stating the hypotheses in terms of p or π .		
	M1	For sight of the correct model $B(60, 0.4)$. May be implied by a correct final answer or a correct probability value: awrt 0.956 or 0.0445		
	A1	awrt 0.0445 (allow awrt 0.044 if correct model seen). Allow CR $X \geq 33$ [and $X \leq 16$]		
	A1	Correct conclusion. Can be awarded even if the candidate has not scored the B1, however, their conclusion cannot contradict their hypotheses i.e if they’ve stated a 1-tail test, then H_0 must be rejected.		
(b)	B1ft	Multiplying their answer in (a) by 2 as long as p -value ≤ 0.5		

Question		Scheme	Marks	AOs
3(a)		0.82	B1	1.1b
			(1)	
(b)		$25 + \frac{21}{31} \times 1 = \dots$	M1	2.1
		$= 25.677419\dots$ awrt 25.7	A1	1.1b
			(2)	
(c)(i) (ii)		$\bar{x} = \frac{2560}{100} = 25.6$	B1	1.1b
		$\sigma_x = \sqrt{\frac{65751}{100} - ("25.6")^2}$	M1	1.1b
		$= 1.46628\dots$ awrt 1.47	A1	1.1b
			(3)	
(Total 6 marks)				
Notes:				
(a)	B1	0.82 oe		
(b)	M1	Correct method to find the median. Allow use of $(n + 1)$ rather than n , leading to 25.6935		
	A1	awrt 25.7		
(c)(i) (ii)	B1	For correct value seen.		
	M1	Correct method to find the standard deviation or variance, follow through their mean.		
	A1	awrt 1.47		

Question	Scheme					Marks	AOs										
4(a)	P (not square number) × P (even number) = $\frac{3}{4} \times \frac{1}{2}$ (o.e)					M1	1.1b										
	$= \frac{3}{8}$ *					A1*	2.1										
						(2)											
(b)	<table border="1"><tr><td>s</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>P(S = s)</td><td>$\frac{1}{4}$</td><td>$\frac{3}{8}$</td><td>$\frac{3}{16}$</td><td>$\frac{3}{16}$</td></tr></table>					s	1	2	3	4	P(S = s)	$\frac{1}{4}$	$\frac{3}{8}$	$\frac{3}{16}$	$\frac{3}{16}$	B1 B1	2.1 1.1b
	s	1	2	3	4												
	P(S = s)	$\frac{1}{4}$	$\frac{3}{8}$	$\frac{3}{16}$	$\frac{3}{16}$												
					(2)												
(c)	$\frac{3}{8} + \frac{3}{16} = \frac{9}{16}$					B1ft	1.1b										
						(1)											
(Total 5 marks)																	
Notes:																	
(a)	M1	Correct method for finding P(S = 2)															
	A1*	$\frac{3}{8}$															
(b)	B1	At least one correct probability for S = 3 or S = 4															
	B1	Fully correct table. Accept different letters for S.															
(c)	B1ft	For $\frac{3}{8}$ + their probability for S = 3 evaluated correctly, ft their $P(S = 3) < \frac{5}{8}$															

Question	Scheme		Marks	AOs
5(a)	The box plot should have the box towards the left as the raw data has a lot of zeroes/low data. or The outliers should be the higher values, not the lower values.		B1	2.2b
			(1)	
(b)	$x + 1.5(x - 1000) < 3400$ or $x + 1.5(x - 1000) \geq 3300$		B1	2.1
	$x + 1.5x - 1500 < 3400$ $x < 1960$		M1	1.1b
	$x + 1.5x - 1500 \geq 3300$ $x \geq 1920$		M1	1.1b
	$1920 \leq x < 1960$		A1	2.2a
			(4)	
(c)	This would lead to the higher values being <u>additional</u> outliers...		B1	3.2a
	...as $1400 + (1400 - 1000) \times 1.5 = 2000$ and the higher end of the data all lie above 2000 making them outliers too.		B1	2.2a
			(2)	
(d)	Quota sampling		B1	1.2
			(1)	
(e)	Might not be enough data for <u>fresh</u> or <u>moderate</u>		B1	2.2b
			(1)	
(Total 9 marks)				
Notes:				
(a)	B1	Statement that the real data is weighted towards the lower values.		
(b)	B1	Correct statement of outliers $UQ + 1.5(IQR)$ or $LQ - 1.5(IQR)$ with the correct sign according to the value used (3400, 3300, 400, 600).		
	M1	Correct process for finding the upper boundary of the upper quartile.		
	M1	Correct process for finding the lower boundary of the upper quartile.		
	A1	Selecting the correct boundaries and showing them in a correct inequality. Accept $1920 < x < 1960$		
(c)	B1	Statement indicating that this would lead to an increase in the number of outliers		
	B1	Calculation leading to 2000 being the upper limit for outliers would mean all of the listed values of 3300 – 3700 (as well as all those above 2000 that are unlisted) would be outliers too		
(d)	B1	Quota sampling		
(e)	B1	Not likely to meet the fresh or moderate quota.		

